

ProjectHub — Full Testing Checklist

Run these in order. Each section depends on the one before it, so if something in Section 3 fails, check Sections 1-2 passed cleanly first before assuming Section 3's code is broken.

General technique for every section:

- Keep browser DevTools open (F12 → Console tab) the whole time — a silent JS error there often explains a UI that "just doesn't work."
- When a test result is ambiguous, check the actual data in Supabase's Table Editor rather than guessing from the UI.
- Use two browser windows (one normal, one Incognito) logged in as two different test accounts for anything involving two people.

0. Environment sanity (5 min)

- ☐ Vercel → Settings → Environment Variables has all 8:
NEXT_PUBLIC_SUPABASE_URL, NEXT_PUBLIC_SUPABASE_ANON_KEY,
SUPABASE_SERVICE_ROLE_KEY, ALLOWED_EMAIL_DOMAIN, RESEND_API_KEY,
NEXT_PUBLIC_VAPID_PUBLIC_KEY, VAPID_PRIVATE_KEY, CRON_SECRET
- ☐ Redeploy once after confirming, so the latest code + all env vars are live together
- ☐ Visit the live Vercel URL (not localhost) and confirm the landing page loads

1. Auth (10 min)

- ☐ Sign up a new student with a @yourdomain email → lands on /home
- ☐ Sign up attempt with a non-matching domain email → correctly rejected
- ☐ Log out, log back in as that student → works
- ☐ Log in as HoD → lands on /hod/dashboard, shows gold "HoD" identity
- ☐ Try logging into HoD Login with a student's credentials → correctly rejected

2. Teams (15 min) — needs 2 student accounts

- ☐ Student A creates a group
- ☐ Student A searches for Student B, sends invite

- [] Student B sees the invite on /team **and** as a bell notification with Accept/Decline buttons
- [] Accept from the **bell notification directly** (not the /team page) → Student B is now a member
- [] Confirm Student A got a notification that B accepted
- [] Try inviting a 3rd already-grouped student → correctly blocked (they shouldn't appear in search)

3. Topics (15 min)

- [] Browse topics as Student A → see available topics with correct categories
- [] Request one → status flips to Pending, HoD gets notified (bell + email)
- [] Propose a custom topic as a **different, ungrouped-yet** test scenario → same HoD notification fires
- [] **Race condition check (if you want to be thorough):** open the same topic in two browser tabs as two different groups, click Request in both quickly — only one should succeed, the other should show "someone just took this"

4. Approvals (10 min)

- [] Approve a pending request from the **Approvals page** → topic locks, group roster freezes, 4 milestones appear on /project
- [] Reject a different request from the **bell notification directly** → topic returns to Available
- [] Request Changes on a third → group sees pending_group_review status
- []  **Known gap:** there's currently no UI for the group to resubmit after Request Changes — don't spend time testing a fix for this, it's expected to be stuck right now

5. Progress & Submissions (15 min)

- [] /project shows correct % and milestone list for the approved topic
- [] Manually mark Frontend/Backend done → % updates, "Mark Done" buttons work
- [] Submit a synopsis PDF → uploads, Synopsis milestone auto-completes, HoD gets notified
- [] Submit final project (GitHub + deployed link) → Final Submission milestone auto-completes
- [] Confirm 100% triggers the confetti/congratulations state

- [] As HoD, open Groups → expand the group → leave feedback on a submission → confirm it appears on the student's /submission page
- [] In HoD Settings, set deadline to yesterday, save → submit something new as student → confirm the red late warning shows and it saves with a "Late" tag

6. HoD tools (10 min)

- [] Dashboard numbers match what you know is actually in the DB
- [] Add a new topic via Projects → shows up on student Browse immediately
- [] Edit and delete a topic (delete should refuse if locked)
- [] Groups page shows correct member lists and progress %
- [] Post an announcement → shows on student /home feed and triggers notifications

7. Notifications — all 3 channels (15 min)

- [] **In-app bell:** unread count is accurate, clears on open, updates within 30s of a new event
- [] **Email:** check a real inbox (not the fake @yourdomain address) receives it — see earlier notes on temporarily switching ALLOWED_EMAIL_DOMAIN to test this properly
- [] **Push:** on the **live Vercel URL** (not localhost), go to Profile → Enable Notifications → allow permission → trigger any event from another account → confirm an OS-level notification banner appears
- [] Trigger the cron manually (Vercel → Cron Jobs → Run) and check the response shows sensible warned/expired counts

8. Chat (10 min) — needs 2 accounts open simultaneously

- [] Group chat: message sent in Window A appears **instantly** in Window B with no refresh
- [] HoD DM: same instant-appearance test between a student and HoD
- [] Confirm a student in a **different** group cannot see this group's messages (check by logging in as them and trying /chat/group — they should only ever see their own group's thread)

9. RLS negative tests (10 min) — confirming things correctly FAIL

These should all be **refused**. If any of these succeed, that's a real bug to report back.

- [] A student tries to edit `settings` or `topic_categories` directly (via browser console: `supabase.from('settings').update(...)`) → should fail
- [] A student tries to read another group's `group_invites` rows → should return empty
- [] A student tries to insert into `topic_status_history` directly → should fail (HoD-only)

If something fails

Note: which section, which exact step, and paste the exact error text (browser console error, or the on-screen message). That's enough for a fast, targeted fix instead of a guessing game.